

Purifying selection on the IP₃ receptor is strongest at the gate and filter and weakest in a luminal loop beside them

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Abstract

The inositol 1,4,5-trisphosphate receptor (IP₃R) is the calcium-release channel of the endoplasmic reticulum, and vertebrates carry three paralogues, ITPR1, ITPR2 and ITPR3, whose missense variants cause ataxias, anhidrosis and neuropathy. Where along its 2,700 residues purifying selection acts has not been measured on alignments deep enough to score a residue. We fitted codon models to 57 vertebrate coding sequences, each proved to encode the protein it was aligned as, and computed per-residue conservation on four nested alignment layers, the deepest holding 249 to 265 genome-derived orthologues of each human paralogue. All three paralogues are under strong purifying selection (one-ratio ω 0.0238 to 0.0430), and ITPR1 is the most constrained and is intensified relative to its sisters (RELAX $k = 9.36$). Along the protein, the gate and the selectivity filter are the most conserved elements in all three paralogues on a composition-free metric, and the gate is identical between all three human copies. A 50-residue luminal loop inside the pore domain, located from the structure's membrane geometry rather than from any annotation, is the least conserved sequence in the receptor; left inside the pore domain it makes the pore look less constrained than the linkers. Scored as a classifier of 1,753 ClinVar missense records, conservation separates pathogenic from benign positions best on the family-wide layer (AUC 0.872), and the deep within-paralogue layer is third of four. Because 1,546 records (88 %) are of uncertain significance, we release every residue's scores as a stratification resource and do not use it to call variants.

Introduction

The inositol 1,4,5-trisphosphate receptor is a tetrameric calcium-release channel of about 2,700 residues per subunit, gated by IP₃ and calcium at a cytosolic ligand-binding region and conducting through a pore formed by the last six transmembrane helices of each subunit [1, 2, 3]. Cryo-electron microscopy has located its working parts at residue resolution: the ten residues within 4.5 Å of bound IP₃, the selectivity filter, and the hydrophobic gate at the cytoplasmic end of the pore [4, 3, 5]. Vertebrates carry three paralogues. Missense variants in *ITPR1* cause spinocerebellar ataxias 15 and 29 and Gillespie syndrome [6, 7, 8, 9], loss of *ITPR2* function abolishes sweating [10], and *ITPR3* variants cause a demyelinating neuropathy and, in one recurrent case, a multisystem disorder [11, 12, 13].

Two kinds of evidence say how strongly a protein is held to its sequence, and they are not the same measurement. A codon model estimates the ratio of non-synonymous to synonymous substitution rates on a tree [14].

It measures selection directly, but it needs coding sequences that provably encode the protein aligned, and it degrades where synonymous sites saturate. Per-residue conservation measures how dispersed a column of an alignment is [15, 16]. It needs no synonymous signal, but it needs many sequences per column, and a column of a 2,700-residue protein cannot be scored from the dozen or so orthologues a representative phylogenetic alignment holds per paralogue. Population constraint from human variation [17] answers a third question, about the last few hundred thousand years, and is not what we measure here.

This paper asks where along the IP₃ receptor purifying selection acts. We answer it on both axes and keep them apart. On the rate axis we fit one-ratio, branch, branch-site and site models and a relaxation test to the three vertebrate paralogues, and carry per-site rates onto the human proteins. On the conservation axis we build orthologue sets deep enough to score a residue, from one locus per genome per paralogue in a 309-genome vertebrate sweep, and compare them against a shallow layer that shows what the representative alignment alone would have given. We then place every element of the receptor against the same protein’s own linkers, test the measured functional residues against their own elements, and ask whether the result classifies the family’s clinical missense variants.

Three things make the answer less obvious than it looks. The Pfam signature named after the ligand, PF08709, contains none of the ten measured IP₃ contacts, which sit in MIR and RIH, the domains the family shares with the ryanodine receptors [18, 19, 20]. The pore domain contains a stretch that no annotation carries and that behaves unlike the rest of it. And the clinical record is mostly uncertain, so a classifier trained on it is scored on very few positions. Each is reported with the control that measures it.

Results

All three paralogues are under strong purifying selection

We recovered a coding sequence for all 57 vertebrate IP₃ receptor tips of the representative alignment, 43 from protein-database cross-references and 14 from genome gene models, and accepted each only if its translation reproduced the aligned protein; 24 codons that disagreed were masked rather than kept (Supplementary Fig. 1). The trimmed codon alignment holds 2,459 codons. One-ratio models give $\omega = 0.02375$ for ITPR1, 0.04303 for ITPR2 and 0.04152 for ITPR3 (Fig. 1; Table 1). The least constrained paralogue therefore accumulates about one non-synonymous substitution for every 23 synonymous ones across a 2,700-residue channel. Removing every genome gene model moves no estimate by more than 0.0112 (curated-only ω 0.02118, 0.0318 and 0.03714), so the rates are not an artefact of the reconstructed models.

Synonymous sites are saturated inside a single paralogue. Across pairwise comparisons within a paralogue, 161 of 171 ITPR1 pairs, and 373 of 420 pairs over all three, exceed a synonymous distance of 1.5, with median dS of 5.721, 4.53 and 13.483 for ITPR1, ITPR2 and ITPR3 (Extended Data Fig. 1a). A single paralogue set spans sharks, teleosts and mammals, so saturation is reached within the copies and not only between them. Every rate we report is therefore from a tree-based model, and the pairwise matrix is used only as this diagnostic.

	ITPR1	ITPR2	ITPR3
one-ratio ω	0.02375	0.04303	0.04152
one-ratio ω , curated sequences only	0.02118	0.0318	0.03714
RELAX k (test clade against the other two)	9.3587	0.9085	0.8356
branch-site ω_2 on the stem	5.52937	999 (at bound)	999 (at bound)
share of sites in the ω_2 class	0.1102	0.14149	0.278

Table 1. Rates of the three paralogues, read from the committed codon-model tables. Two stems carry an ω_2 at

the optimiser's upper bound, which is not an estimate.

ITPR1 is the most constrained paralogue and its selection is intensified

Ranked from most to least constrained, the order is ITPR1, ITPR3, ITPR2. Two-ratio branch models, each paralogue clade against the rest of the family, separate every clade from its background after correction across the whole family of 12 likelihood-ratio tests: ITPR1's foreground ω of 0.0241 against a background of 0.0432 ($q = 2.9 \times 10^{-63}$), ITPR2's 0.0435 against 0.0317 and ITPR3's 0.0455 against 0.0308 (Fig. 2a). A relaxation test, which compares whole ω distributions rather than point estimates [21], gives the same ordering by a different statistic: selection on ITPR1 is intensified relative to the other two ($k = 9.3587$), and relaxed on ITPR2 ($k = 0.9085$) and ITPR3 ($k = 0.8356$) relative to theirs (Fig. 2b). ITPR1 is also the paralogue that carries most of the family's pathogenic missense record.

Only the ITPR1 stem carries a branch-site rate the data determine. Branch-site model A asks whether a class of sites evolved faster than neutrally on the stem of a paralogue, the interval in which a new copy was most free to change [22]. We ran it from four initial ω on every stem, and 3 of the 12 restarts converged below their own nested null, one on every stem and from a different starting value each time (Extended Data Fig. 1b). Each of those is a local optimum a single-start run would have reported. All three stem tests are significant after correction, but only one is interpretable. On ITPR1 the foreground class has $\omega_2 = 5.52937$ on a share of 0.1102 of sites, stable across the restarts that reach the best likelihood, with 8 sites above a posterior of 0.95 [23]. On the ITPR2 and ITPR3 stems ω_2 sits at codeml's bound of 999, and on ITPR2 restarts at the same likelihood place it anywhere from 162 to 999: the stem has no synonymous signal left to normalise a rate against. Those two tests are significant and their effect sizes are not measured.

No site in any paralogue is positively selected. Within each paralogue, M2a never gives its positive-selection class a non-zero share, and M8 fits better than M7 in all three only because it adds a class at exactly $\omega = 1$ on under 1 % of sites, with 0, 0 and 1 sites reaching a posterior of 0.95. A class of sites free to drift is not a class being driven.

The gate and the selectivity filter are the most constrained elements

For per-residue conservation we took one locus per genome per paralogue from the vertebrate genome sweep, giving 264 ITPR1, 249 ITPR2 and 265 ITPR3 orthologues (Supplementary Fig. 2), and scored each column with a sequence-weighted Jensen–Shannon divergence [16, 15]. Every element was tested against the same protein's own linkers, since a whole-protein mean contains the element being tested. The linker means are 0.7256, 0.7209 and 0.7214.

On the modal-residue fraction, which uses no background model, the gate and the selectivity filter are the two most conserved elements in all three paralogues (gate 0.9846, 1.0 and 1.0), with the ITPR1 filter tied with the RIH-associated domain (Fig. 3; Table 2). On the divergence metric the gate ranks first in ITPR2 and ITPR3 and second in ITPR1, and the filter ranks second to fourth. The two are five and seven residues long, so their tests against the linkers reach $p < 0.05$ only in ITPR2 and ITPR3 for the gate (0.0154, 0.0224) and in ITPR3 for the filter (0.0237). The most constrained whole domain is the RIH-associated domain (0.8037, 0.8016 and 0.7976), and every named element but one sits above its protein's linker mean.

element	ITPR1	ITPR2	ITPR3
gate	0.7994	0.822	0.822
selectivity filter	0.7866	0.7966	0.8114
RIH-associated	0.8037	0.8016	0.7976
pore domain, luminal loop removed	0.7532	0.7497	0.7489
luminal loop	0.4883	0.4521	0.4846

element	ITPR1	ITPR2	ITPR3
linkers (control)	0.7256	0.7209	0.7214

Table 2. Mean Jensen–Shannon divergence per element on the deep layer. Higher is more conserved.

A luminal loop is the least conserved sequence in the receptor

The one exception sits inside the pore domain. We located it from the structure rather than from any annotation, as the contiguous run of channel residues whose C α lies beyond the membrane on the luminal side of the axial span measured on 6DQN [3]: residues 2401 to 2450 of ITPR3, transferred by alignment onto ITPR1 and ITPR2, where 49 and 40 of its residues carry a score. It scores far below every other element and below the linkers in all three paralogues (0.4883, 0.4521 and 0.4846), and its modal-residue fraction (0.5468, 0.5632 and 0.5236) says the same without a background model. On the constraint profile along the pore-forming half, the curve collapses in the loop and recovers on both sides of it (Fig. 4).

Leaving the loop inside the pore domain inverts the pore’s result. With the loop included, the pore domain read as the least constrained named element in ITPR1, below the linkers (JSD 0.697, $p = 0.96$), a strange result for the pore of an ion channel. We checked three explanations. A metric artefact was ruled out by the composition-free metric, which gave the same picture. A gene-model artefact was ruled out by recomputing on the curated sequences alone, where the loop still scored lowest (0.6191 in ITPR1). The third explanation held: the domain’s mean averaged the most and least conserved stretches of the receptor. With the loop separated, the pore domain clears the linker control in all three paralogues ($p = 0.0054$, 0.0011 and 0.0083). Because the loop was drawn on the membrane’s geometry, that result is not the profile explaining itself.

The measured functional residues are conserved beyond their own domains

The ten residues within 4.5 Å of IP₃ in 6DQN all fall in MIR and the N-terminal RIH domain, and none in PF08709, the Pfam signature named after the ligand. Tested against the whole protein, they are more constrained in all three paralogues ($p = 0.013$, 0.0006 and 0.0042). Tested against the rest of the domains that carry them, the harder control, they are more constrained in ITPR2 ($p = 0.0021$) and ITPR3 ($p = 0.0253$) and higher but not significant in ITPR1 ($p = 0.0697$) (Fig. 5a). The filter-lining and gate-lining sets are two residues each, so we report their means and leave their tests to the element level, where those elements are powered.

The gate has not changed in any paralogue since the duplications. Identity between the human paralogues, which needs no alignment depth and no conservation metric, is 1.0 across the five gate residues in all three pairs (Fig. 5b). The luminal loop retains 0.1304, 0.2041 and 0.3111 identity, against 0.6405, 0.6628 and 0.7026 for the whole protein. The two extremes of divergence between the copies therefore lie about fifty residues apart in the same domain. Measured on the trimmed representative alignment instead, the whole-protein identities are 0.7526, 0.7728 and 0.812, which is the difference trimming makes: it keeps 12 of the luminal loop’s 51 residues.

Per-site rates agree on the same coordinates. Carried onto the human proteins, FEL [24] calls 5,766 sites purifying across the three paralogues and one site diversifying, which is inside the procedure’s own false discovery budget. Every gate, filter and contact site in ITPR1 and ITPR3 is called purifying; in the luminal loop the share falls to 0.2667, 0.3333 and 0.2667 of the codons the trimmed alignment kept, against 0.8603, 0.712 and 0.7733 over the whole protein, with median non-synonymous rates of 0.3275, 0.3902 and 0.3946 where elsewhere they are zero. Painted into the B-factor column of 16 experimental and predicted structures, both layers show the same pattern on the channel (Supplementary Fig. 3).

Conservation separates pathogenic from benign variants, and depth is not the best layer

We placed 1,753 ClinVar missense records on the three canonical sequences after checking each cited transcript's own translation against the canonical residue by residue; none was dropped (1,006 ITPR1, 299 ITPR2 and 448 ITPR3 records). A positive control the harvest could have failed passed: both residue-level variants the literature localises, ITPR3 p.Thr1424Met and p.Arg2524Cys [11, 13], were recovered. The record is mostly uncertain: 1,546 records are of uncertain significance (875, 261 and 410), and ITPR2's entire pathogenic record is one position.

The four layers score slightly different sets of residues, so ranking them on their own coverage would compare the sets as much as the layers. We therefore scored them on the one fixed set that every layer scores, 44 pathogenic and 34 benign positions (Fig. 6a). The family-wide layer of 128 IP₃ receptor tips is the best classifier (AUC 0.8723), then the vertebrate layer (0.8536), then the deep within-paralogue layer (0.758), and the shallow control is worst (0.6842). Depth was worth building, since the shallow layer is what the representative alignment alone would have given. Breadth was worth more: a position conserved across the family's eukaryotic range and 500 million years of paralogue divergence discriminates better than one invariant across 260 orthologues of a single gene.

The ranking holds inside the genes that can be scored. Without the restriction to shared positions, the deep layer separates 45 pathogenic from 42 benign positions at AUC 0.7926, higher than on the fixed set, which is why the restriction matters: each layer's own coverage flatters it differently. Within ITPR1, which holds most of the labelled positions, the family layer reaches 0.9013 on the shared set, and within ITPR3 it reaches 0.8533 on five pathogenic positions. We report no per-gene score for ITPR2, whose single pathogenic position cannot define a curve.

The whole-protein control separates two effects. Each layer also separates pathogenic positions from the average residue, but by less (0.7574 for the family layer, against 0.8723 for the benign contrast). Part of the signal is therefore benign calls concentrating in unconstrained positions, not only pathogenic calls concentrating in constrained ones. Pathogenic positions are enriched in the gating elements: 2 of ITPR1's pathogenic positions fall in its five gate residues (odds ratio 48.9, $p = 0.0019$) (Supplementary Fig. 4). ITPR1's pathogenic positions cannot be placed on a structure at all, because neither its rat cryo-electron microscopy reference nor the AlphaFold DB isoform model carries the human residue at every shared position.

Uncertain variants can be stratified, not called. Against the labelled distributions of its own gene on the family layer, 58 of 570 scored ITPR1 variants of uncertain significance, 11 of 196 in ITPR2 and 34 of 299 in ITPR3 sit at or above that gene's pathogenic median (Fig. 6b). That says where a variant sits on an axis the labelled variants separate on. It is not a pathogenicity call, and the ITPR2 row rests on a single pathogenic position.

Discussion

Purifying selection acts on the whole IP₃ receptor, and its strength is not uniform along it. Two independent axes say where it is strongest. On the rate axis, a whole-gene ω of 0.02 to 0.04 in every paralogue and a per-site screen finding thousands of purifying sites and no positively selected one put the receptor among strongly constrained proteins. On the conservation axis, the gate and the selectivity filter are the most conserved elements on a composition-free metric, and the gate is the one element of the channel that all three paralogues have kept identical since the vertebrate genome duplications (companion paper 2). The weakest point lies about fifty residues away, in a luminal loop of the same Pfam domain. The structural literature places the filter and the gate at the heart of conduction and gating [4, 3, 5], and the conservation pattern agrees with it without having been told where those elements are: the loop was located from the membrane geometry, and the filter and gate by transferred structural coordinates with anchor tests that could have failed.

The loop matters beyond its own score. A per-domain summary is the obvious way to read a constraint map, and for this channel it gives the wrong answer: with the loop left in, the pore domain of an ion channel reads as less constrained than the linkers between its cytosolic domains. Any analysis that summarises the receptor by Pfam domain will reproduce that artefact, and the same boundary affects any comparison that uses the pore domain as a reference. The loop is also where the maps of cryo-electron microscopy are weakest, which is why we write unscored residues as missing rather than as unconstrained on the painted structures.

The three copies have not been held to the same standard. ITPR1 is the most constrained on one-ratio, two-ratio and relaxation tests alike, and its stem is the only one whose branch-site rate is identifiable, with a class of about a tenth of sites evolving faster than neutrally on the branch where the copy was new. ITPR1 also carries almost all of the family's pathogenic missense record [6, 7, 8, 9]. We do not read the order of the other two as settled. ITPR2 and ITPR3 differ in one-ratio ω by 0.0015, and ITPR2's near-empty clinical record is better explained by ascertainment than by tolerance, since its gate and IP₃ contacts are as conserved as ITPR1's.

For variant interpretation the practical result runs against the design of the analysis. The deep within-paralogue orthologue sets, which took a 309-genome sweep to build, classify pathogenic against benign positions better than the shallow control but worse than the family-wide and vertebrate layers. Conservation across the family's full range, spanning paralogue divergence and the non-vertebrate grade, is the more informative prior. A resource for this family should therefore quote the family-wide layer first, and population constraint [17] would be a separate, complementary axis rather than a replacement.

Several limits bound these conclusions. Synonymous sites are saturated within every paralogue, so every ω is a tree-based estimate with a soft denominator, and the two stems whose branch-site rate sits at the optimiser's bound are unmeasured rather than large. The codon models condition on the tree and its placement of the paralogue stems (companion paper 2); a branch test cannot corroborate the topology it is run on. The rate analyses are vertebrate-only, and nothing here bears on the single-copy receptors outside the vertebrates (companion paper 1). The deep layer is about 95 % genome translations; the curated-only recomputation leaves the element ranking intact but cannot rule out every gene-model error. The classifier is scored on 44 pathogenic and 34 benign positions, most of them in ITPR1, so every AUC describes that set rather than the family's variant space. Conservation and ω measure different things, a column's dispersion and a substitution-rate ratio, and we report them side by side without merging them. Finally, the stratification of uncertain variants is not a classification, and should not be used as one.

Methods

Scope

The rate analyses cover the 57 vertebrate IP₃ receptor tips of the 134-sequence representative alignment, which carries the phylogeny of the family (companion paper 2); the six ryanodine receptor outgroup tips are excluded because synonymous sites do not survive that distance. The conservation analyses cover the three human paralogues (UniProt Q14643, Q14571 and Q14573 [25]) in their own numbering, on four nested layers: a deep layer of 264, 249 and 265 orthologues of each paralogue, one locus per genome from a sweep of 309 vertebrate genome assemblies (companion paper 3); a vertebrate layer of the 57 vertebrate tips; a family-wide layer of all 128 IP₃ receptor tips; and a shallow layer of each paralogue's own representative tips (19, 13 and 19), kept as the control for what the deep sets bought. The ryanodine receptors are excluded from every layer, so no layer scores what the superfamily conserves. The variant analyses cover every ClinVar missense record on the three genes, 1,753 in all.

Coding sequences and the codon alignment

Every tip needs a nucleotide sequence that provably encodes its aligned protein. Candidates came from three routes (Ensembl, UniProt cross-references and genome gene models from the sweep), and each route was treated as a generator of candidates rather than an answer, because a UniProt entry lists every transcript of its gene and for human ITPR1 and ITPR2 the first listed is not the aligned isoform. Each candidate was translated against the aligned protein and every disagreement masked to NNN, not only internal stops; 24 codons were masked across the set. Genome gene models were realigned with miniprot in a mode that keeps frameshift residues [26] and accepted only if they reproduced the swept protein. PAL2NAL [27] built the codon alignment, which was cross-checked nucleotide by nucleotide against an independent in-house protein-to-codon mapping, any disagreement aborting the build. trimAl [28] columns were chosen on the protein and applied codon-aware, whole triplets only, keeping 2,459 codons of 3,253. Twelve constructed negative controls run on every build and check refusal: a frame-shifted sequence, a sequence encoding a different protein, and a non-monophyletic foreground must each be refused.

Selection sets and codon models

The three selection sets are the extended paralogue clades of the maximum likelihood tree (companion paper 2), re-derived from the rooted tree and cross-checked against the committed clade table, because a codeml branch model given a foreground that is not a clade silently marks a larger one. Six cyclostome tips that the tree places in no paralogue clade stay in the whole-tree analyses as background and are in no foreground. With PAML 4 [14] we fitted one-ratio models per paralogue and on curated sequences only, pairwise estimates, a whole-tree one-ratio null, a two-ratio model per paralogue clade, branch-site model A on each paralogue stem [22], and site models M1a/M2a and M7/M8 within each paralogue, with Bayes empirical Bayes posteriors [23]. Branch-site model A was started from $\omega = 0.5, 1.5, 2.5$ and 4 on every stem, and the best restart is reported; the branch-site test is read against a 50:50 mixture of χ^2_0 and χ^2_1 because its null sits on a parameter boundary. Benjamini–Hochberg correction [29] runs across all 12 likelihood-ratio tests together. Pairwise synonymous distances above 1.5 are flagged as saturated. RELAX [21] in HyPhy [30] tested each paralogue clade against the other two, leaving unplaced tips unlabelled rather than in the reference.

Per-site selection

FEL [24] was run on each paralogue’s codon alignment and every site carried onto the human protein by re-alignment; a site whose amino acid disagreed after transfer is written with no residue. Because synonymous sites are saturated, we report the median non-synonymous rate and the share of sites called purifying at $q < 0.05$ per element, and take ω only over sites whose synonymous rate is identifiable. That rate is unidentifiable at 671 of 7,368 scored sites.

Orthologue sets and conservation

Deep orthologues were taken from the sweep’s gene models at bait coverage ≥ 0.8 , identity ≥ 0.4 and at least 1,800 aligned residues, one per genome and paralogue, with lesion-rich loci excluded. Because bait coverage cannot see a model that is too long, each sequence was also scored on the fraction of its own residues landing in reference columns, with the bar placed at the midpoint of the gap in that distribution (0.7133). It removed one chimeric model, which had been inflating the ITPR2 alignment from 3,380 to 5,676 columns. Alignments were built with MAFFT L-INS-i [31] at a single thread for reproducibility. Columns were scored by Jensen–Shannon divergence against the BLOSUM62 background [15] after position-based sequence weighting [16], with gaps treated as missing data and the occupancy of each score reported. A composition-free metric, the modal-residue fraction, and a recomputation on the curated sequences alone are written beside every element, not only where a result surprised.

Elements, controls and functional sites

Pfam elements [32] were taken per accession from InterPro [33] and need no transfer. The selectivity filter, the gate and the ten IP₃ contacts (residues within 4.5 Å of IP₃) were measured on the human ITPR3 structure 6DQN [3] and transferred to ITPR1 and ITPR2 by pairwise alignment, each with an anchor test that aborts rather than writing a coordinate: the filter must arrive on the GGGVGD motif and the gate on the same lining residues. The luminal loop was defined geometrically on the same structure. Residues outside every element were assigned to named linkers and termini, which form the within-protein control; each element is tested against its protein's linkers by a one-sided Mann–Whitney test. Functional residues were tested against the whole protein and against the rest of their own elements. Identity between paralogues was measured over mutually covered columns of the untrimmed alignment, with the trimmed-alignment value carried beside it. Fourteen constructed negative controls run before any table is written, including a mutated filter motif that the anchor must refuse and a domain that must not enter the linker control; they were mutation-tested on four deliberate rule breakages.

Variants

ClinVar missense records were placed only after each cited transcript's own translated coding sequence was aligned to the canonical sequence and the reference amino acid required to match; all three genes file on a transcript whose translation is the canonical, so none was dropped. The two residue-level variants localised in the literature must be recovered, as a positive control. Each layer was scored as a classifier by ROC AUC with a Mann–Whitney test on three contrasts: pathogenic or likely pathogenic against benign or likely benign, pathogenic against the whole protein (the control for ascertainment), and the first contrast restricted to the 44 pathogenic and 34 benign positions every layer scores, which is the contrast the layers are ranked on. Enrichment by element used Fisher's exact test. Variants of uncertain significance were stratified against the pathogenic and benign medians of their own gene, thresholds chosen by the labelled distributions rather than by us.

Statistics and reproducibility

Element and site comparisons are one-sided Mann–Whitney tests against the named control; classifier significance is a Mann–Whitney test on the two labelled sets. Variant classes were collapsed into pathogenic or likely pathogenic, benign or likely benign, uncertain, conflicting and other, and only the first two enter a classifier. No test is reported on fewer than three observations, which is why the two-residue filter-lining and gate-lining sets carry means only. A p-value is written with its significant figures and never rounded to zero. MAFFT runs single-threaded because the L-INS-i result is not reproducible across thread counts, and every alignment, table and figure is recorded with its SHA-256 so that a rebuild which drifts is visible in the data.

Structures

The deep-layer conservation and the inverted FEL rate were written into the B-factor column of each IP₃ receptor structure in the panel, with unscored residues set to (-)1 so that missing data cannot read as low constraint. A chain mapping under half of itself to its paralogue was refused. A structure may carry a human variant position only if every residue it shares with the human table carries the same amino acid.

Figure legends

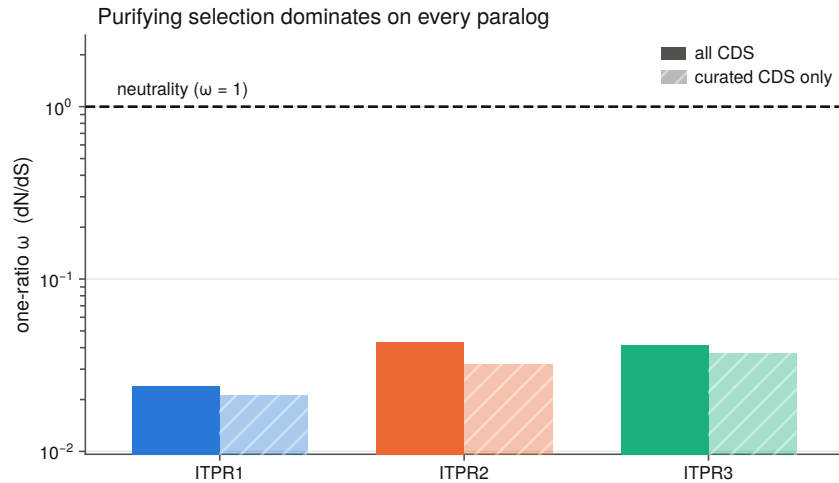


Fig. 1. All three paralogues sit more than twenty-fold below neutrality. Each bar is a paralogue's one-ratio ω from the full codon set, the bar beside it repeats the estimate with every genome gene model removed, and the dashed line marks $\omega = 1$. The axis is logarithmic because the result is the distance below neutrality, which a linear axis would flatten onto the floor. Removing the reconstructed gene models barely moves any bar, which shows that they do not drive the rates.

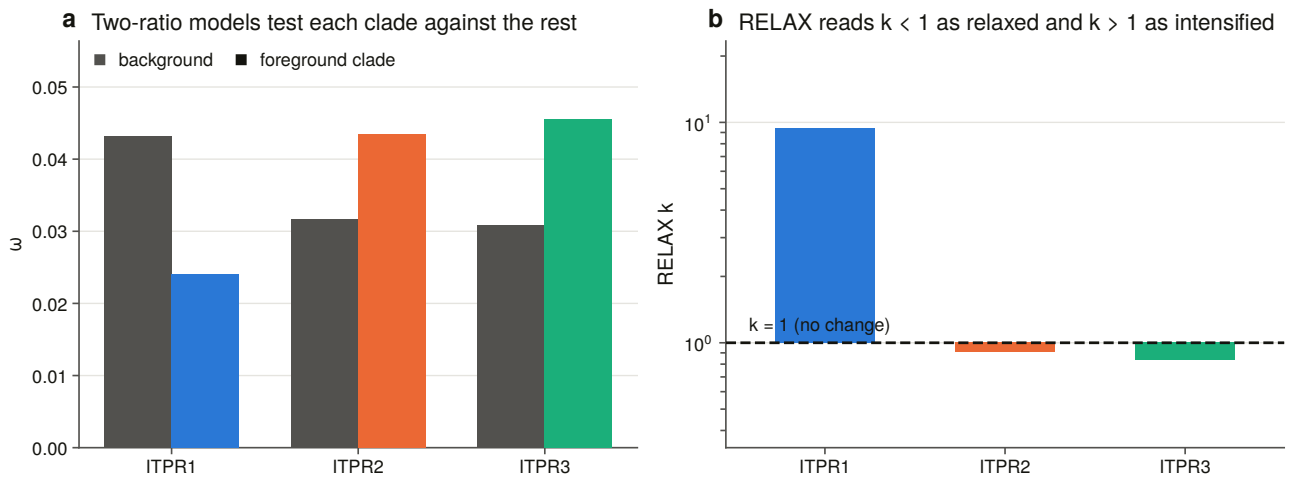


Fig. 2. Two statistics on two models separate ITPR1 from its sisters in the same direction. Panel (a) sets each paralogue clade's foreground ω against the background ω of the rest of the family in a two-ratio model. Panel (b) gives the RELAX selection-intensity parameter k for the same three contrasts, where k above 1 means intensified and below 1 relaxed. A point-estimate contrast and a test on whole rate distributions agree that ITPR1 is held more tightly than ITPR2 and ITPR3.

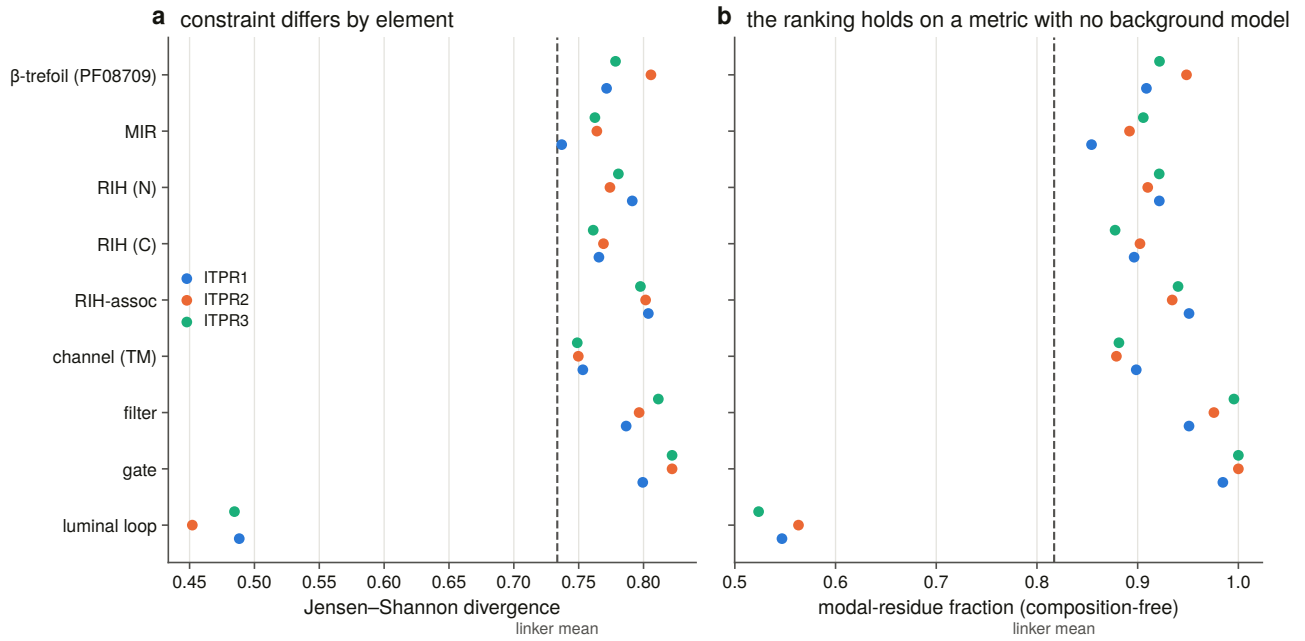


Fig. 3. The gate and the selectivity filter are the most conserved elements of the receptor when composition is left out of the metric, and the luminal loop is the least conserved on both metrics. Panel (a) ranks every element of the three paralogues by mean Jensen-Shannon divergence on the deep layer; panel (b) repeats the ranking on the modal-residue fraction, which uses no background frequencies. Dashed lines give each protein's linker mean, the within-protein control. The two metrics agree on the extremes, so neither is a product of scoring transmembrane residues against a background table.

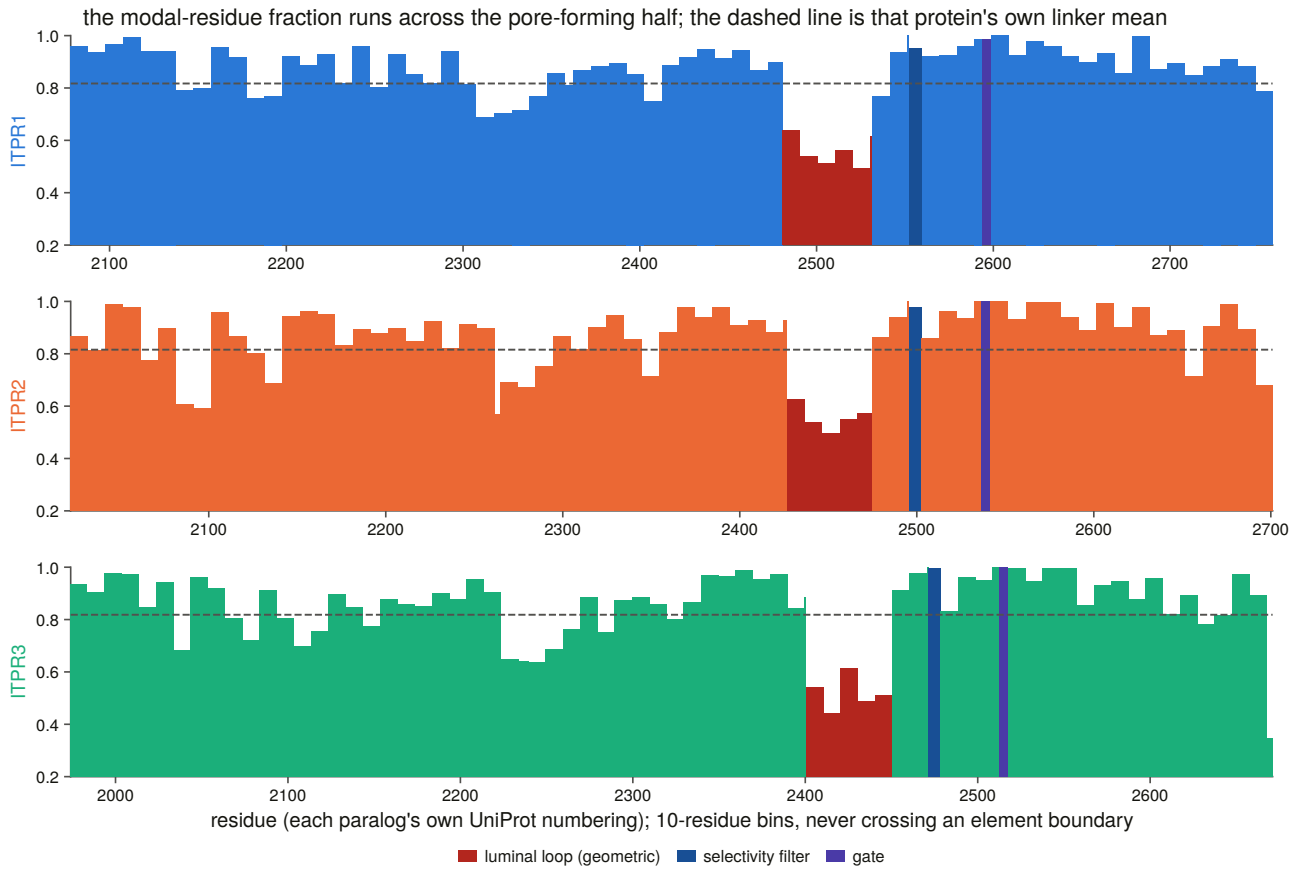


Fig. 4. Conservation collapses in the luminal loop and recovers on both sides of it in all three paralogues. Each row plots the modal-residue fraction along the pore-forming half of one paralogue in its own numbering, in ten-residue bins that never cross an element boundary, so the edge of the loop is drawn where the structure puts it. The loop, the filter and the gate are coloured, and the dashed line is that protein's linker mean. The loop was located from the membrane geometry of 6DQN, so the dip is a prediction the profile confirms rather than a boundary drawn on it.

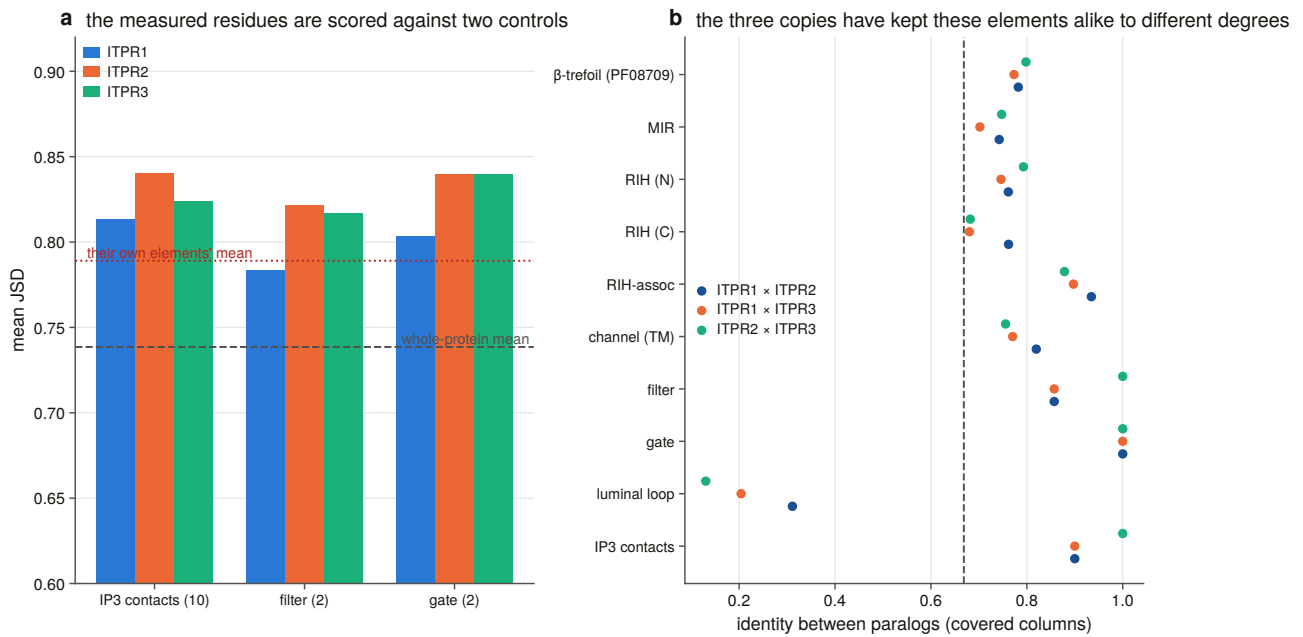


Fig. 5. The measured functional residues clear the easy control and most of them clear the hard one. Panel (a) scores the ten IP₃ contacts, the two filter-lining and the two gate-lining residues against the whole protein and against the rest of the element each sits in. Panel (b) gives identity between each pair of human paralogs for every element, with the gate at full identity in all three pairs and the luminal loop lowest. Beating the rest of its own element is what shows that the structural annotation picks out residues its domain does not already share.

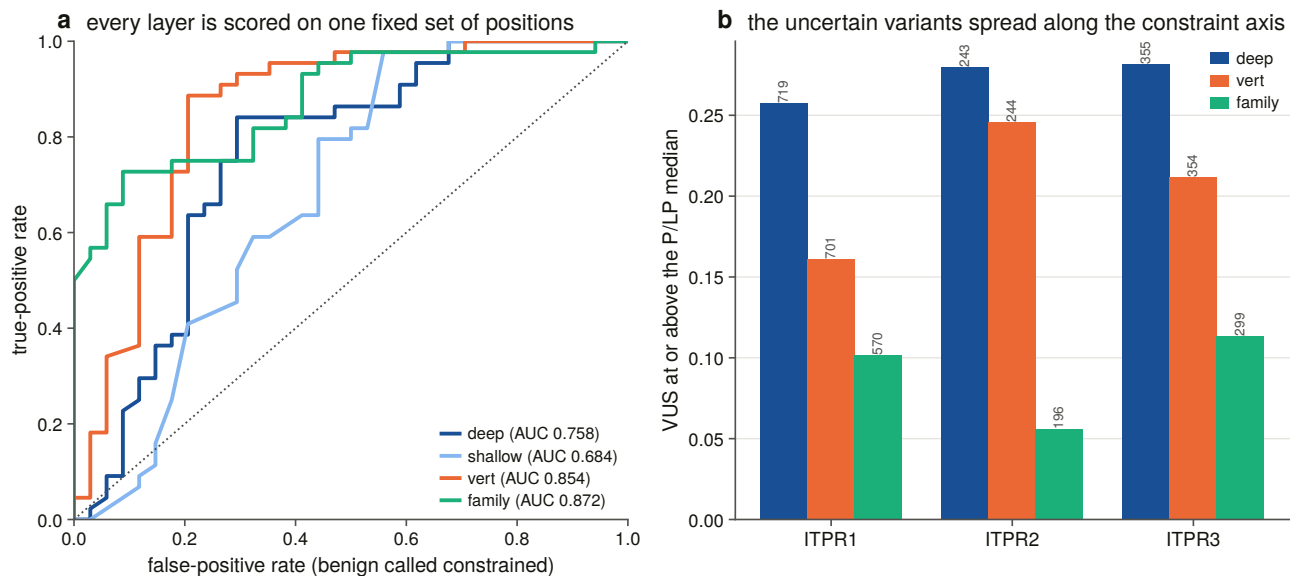
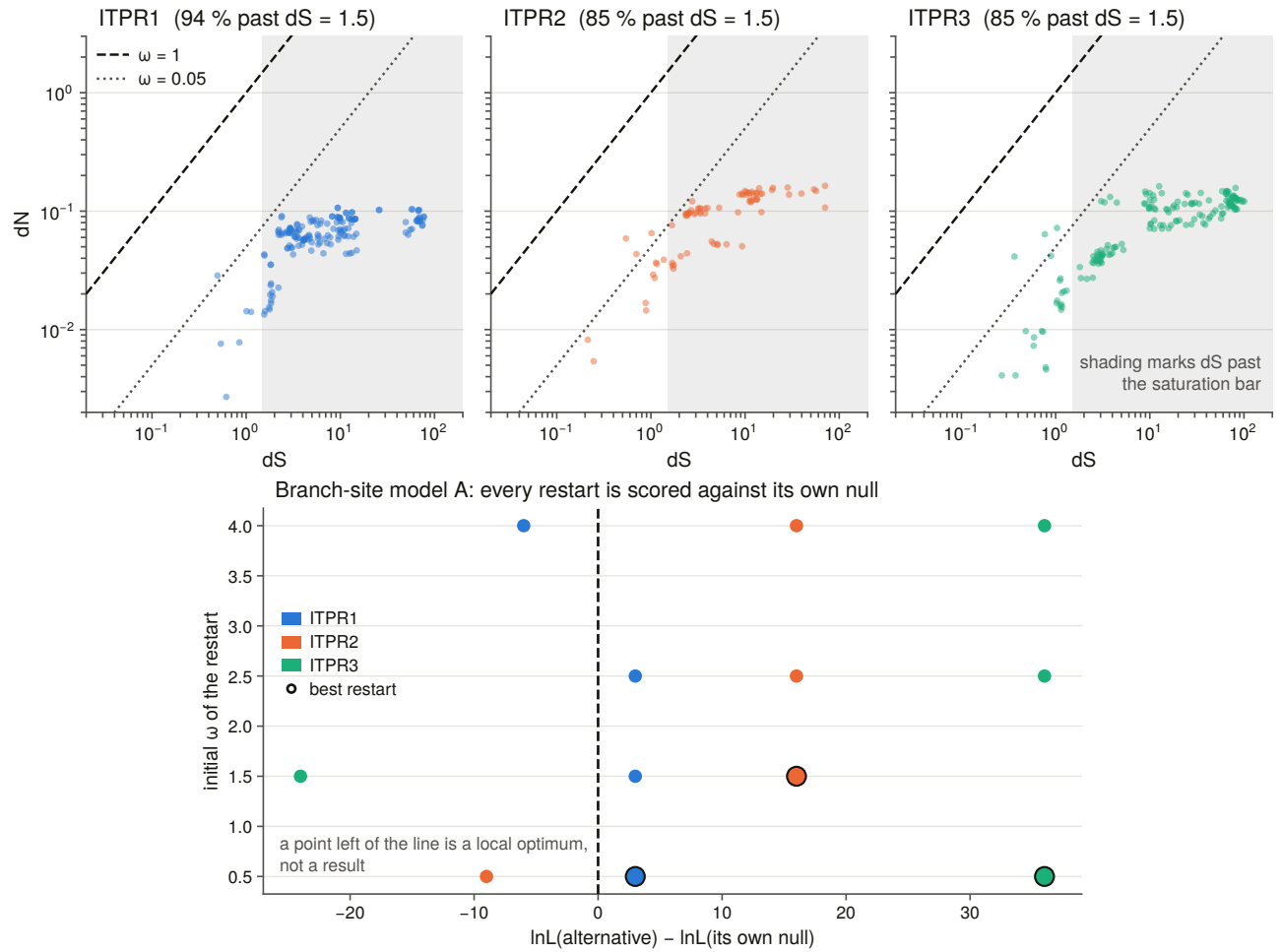
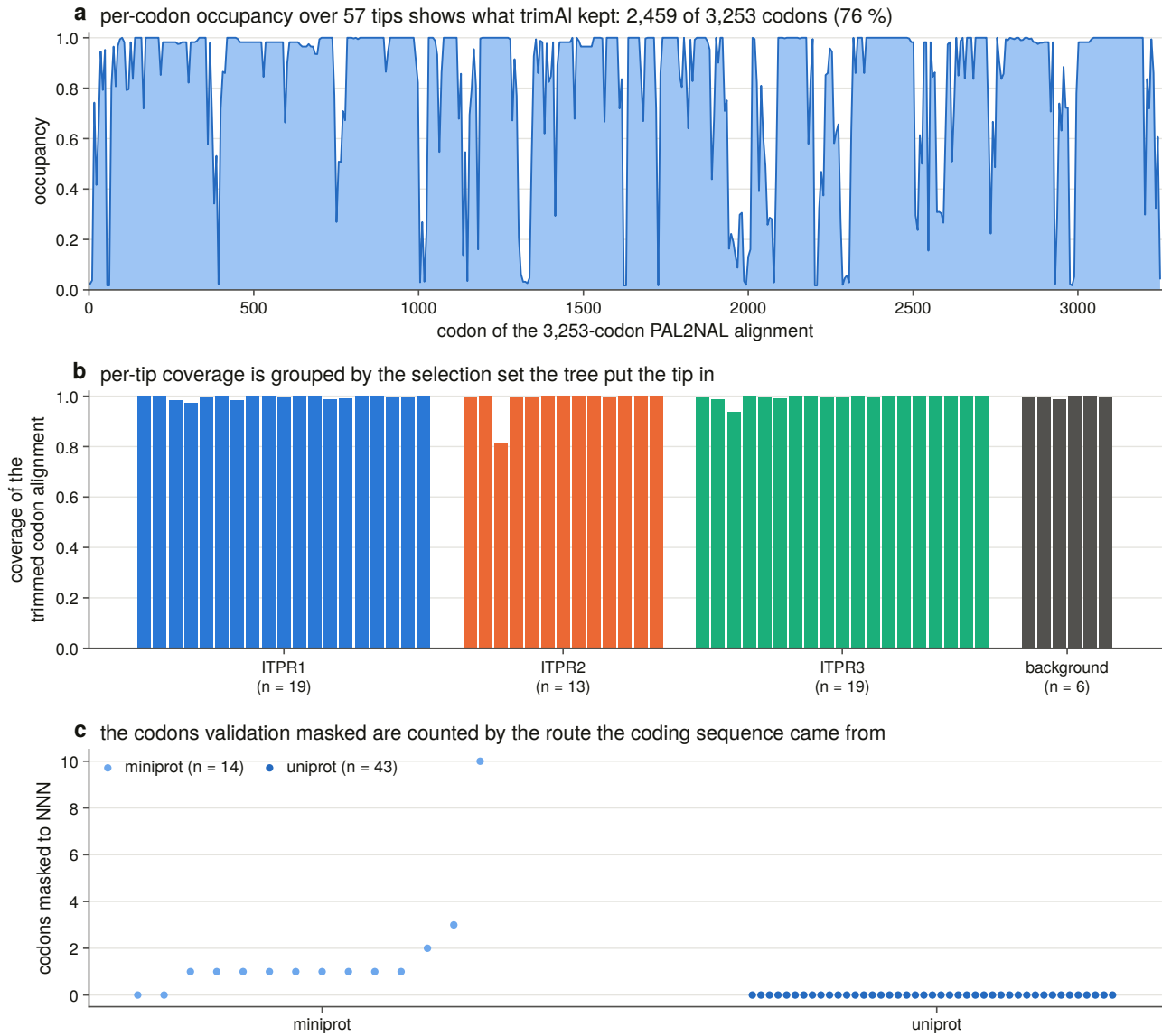


Fig. 6. The family-wide layer classifies clinical variants best, and the deep within-paralogue layer ranks third. Panel (a) draws receiver-operating curves for the four layers on the fixed set of pathogenic and benign positions that every layer scores, as full curves because the layers cross. Panel (b) places the variants of uncertain significance on the same axis against their gene's labelled medians. A resource for this family should quote the layer at the top of panel (a), and panel (b) is a stratification, not a call.

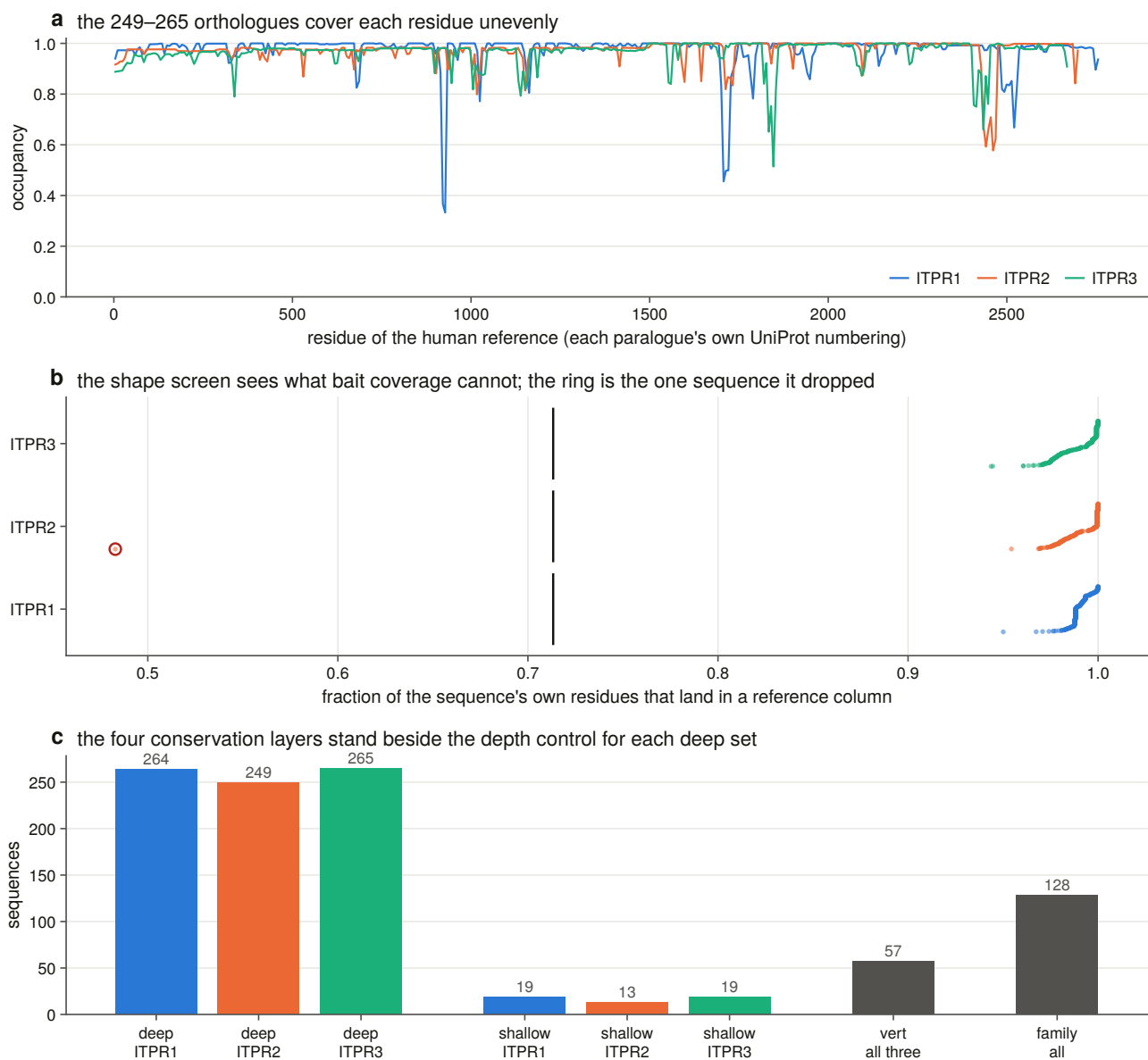
Extended Data



Extended Data Fig. 1. Two checks on the codon models show why every rate here comes from a tree and from the best of several optimiser runs. Panel (a) plots dN against dS for every pair of tips within each paralogue on log–log axes, with the neutral diagonal and the saturation bar at $dS = 1.5$; most pairs lie past the bar in all three paralogues. Panel (b) plots every branch-site restart against its own nested null likelihood, and the three points below the line are local optima that a single-start run would have reported as answers.



Supplementary Fig. 1. The trimmed codon alignment carries every rate in this paper. Panel (a) plots per-codon occupancy across the 57 tips, marking the columns trimming kept; panel (b) gives each tip's coverage grouped by the selection set the tree placed it in; and panel (c) counts the codons that validation masked, by the route each coding sequence came from. The axis is in codons, since a nucleotide axis would show a threefold difference that is not in the data.

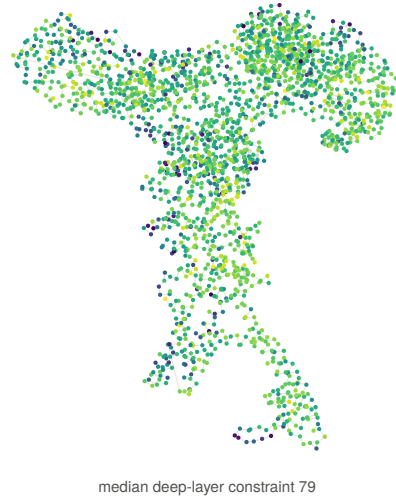


Supplementary Fig. 2. The deep alignments behind the constraint map thin out in places, and the figure shows where. Panel (a) counts how many of the 249 to 265 orthologues cover each residue of the human reference; panel (b) shows the shape screen, with a ring on the single chimeric model it removed; panel (c) sets the four conservation layers beside the depth control. Scores are less reliable where coverage thins than where it is full.

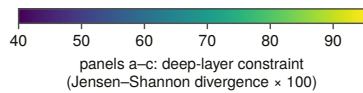
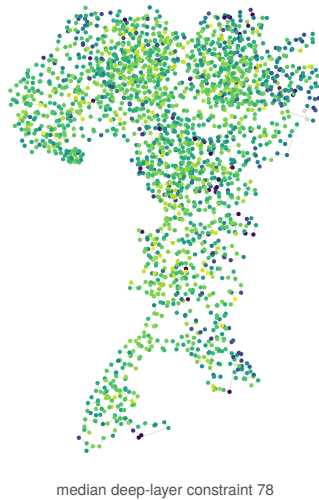
a ITPR1 (7LHF) resolves 2,300 residues



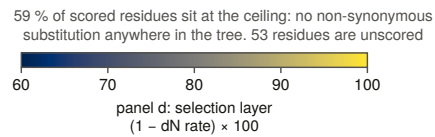
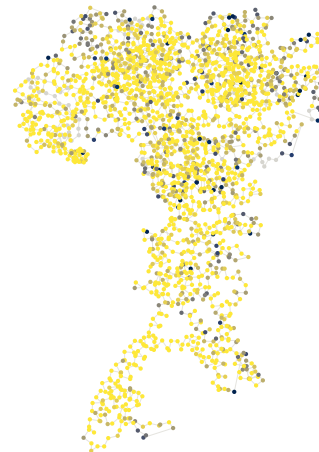
b ITPR2 (9YKK) resolves 2,168 residues



c ITPR3 (8TKG) resolves 2,210 residues

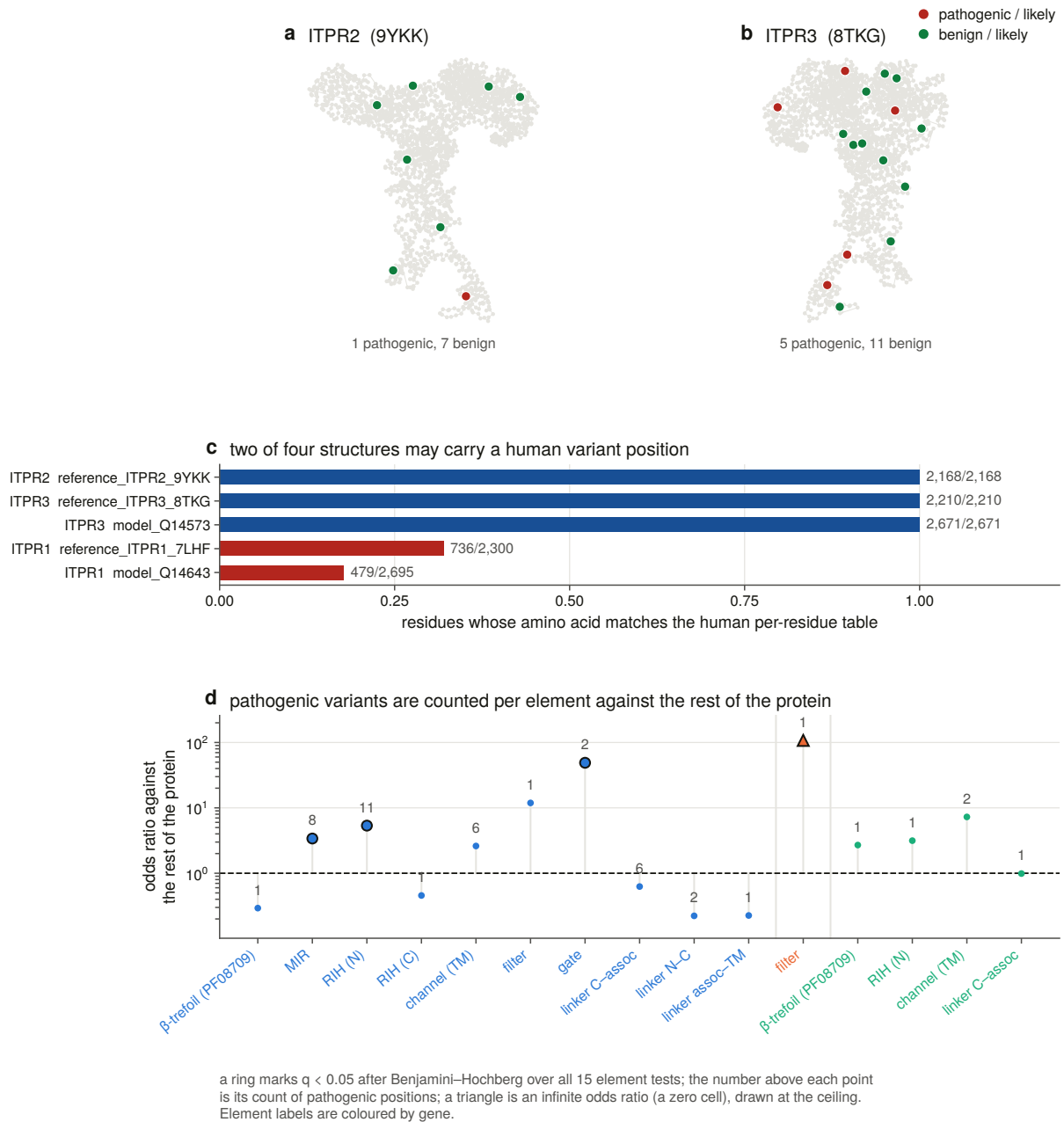


d ITPR3 (8TKG) is painted with the selection layer instead



● residue with no score (written -1 by S17, never 0)

Supplementary Fig. 3. Painting the constraint map onto one experimental structure of each paralogue shows the most conserved residues on the pore and leaves the luminal loop grey. Panels (a) to (c) colour each trace from the B-factor column the painting step wrote, and panel (d) paints the per-site selection layer onto ITPR3. Unscored residues appear in grey rather than at the low end of the scale, which keeps missing data from reading as an absence of constraint in the loop, where the two coincide.



Supplementary Fig. 4. Pathogenic positions concentrate in the elements that gate and select. Panels (a) and (b) place every labelled variant on the ITPR2 and ITPR3 structures allowed to carry a human position; panel (c) records the numbering rule for each candidate structure, which the two ITPR1 structures fail; and panel (d) tests the enrichment of pathogenic positions per element, with a ring where the test survives correction. An infinite odds ratio is drawn at the ceiling with a marker so that the strongest enrichment stays on the axes.

Data availability

Every table, alignment, tree, figure and report this paper rests on is in the deposited archive, listed file by file with its size and SHA-256 in `papers/constraint/deposit_manifest.tsv`. The primary results are `results/selection/` (codon models, RELAX, the codon alignment and its validation) and `results/constraint/` (the per-residue tables, the orthologue sets, the variant tables and the painted structures). The per-residue deliverable is one table per human paralogue, `constraint_<gene>_<accession>.tsv`, giving every residue in its own UniProt numbering with four conservation layers, the occupancy each is conditional on, its element, and whether it is a measured IP₃ contact, filter or gate residue; `variants.tsv` joins the same columns onto every harvested ClinVar record. These scores stratify variants of uncertain significance against the labelled variants of the same gene. They are not pathogenicity calls and should not be used as such. All primary data are public: genome assemblies from NCBI Datasets, protein sequences from UniProt, the structure 6DQN and the painted cryo-electron microscopy entries from the RCSB Protein Data Bank, and variant records from ClinVar.

Code availability

Every analysis is regenerated by the scripts in the repository (`scripts/s9_*.py`, `scripts/s17_*.py`), each of which runs its constructed negative controls before writing anything. This paper is rebuilt by `python scripts/s26_assemble.py --paper constraint`, which re-verifies every load-bearing number in it against the committed tables.

Author contributions

George Dickinson directed the project, chose its questions and reviewed its results. The analysis code and the text of this paper were written by Claude (Anthropic) under his direction; Chapter 16 of the accompanying thesis describes how the work was carried out.

Competing interests

The author declares no competing interests.

Companion papers

1. George Dickinson. The IP₃ receptor is ancestrally eukaryotic and has been lost repeatedly outside the animals. *In preparation* (paper 1 of 6 in this series; source in `papers/range/`).
2. George Dickinson. The three vertebrate IP₃ receptors arose in two separate early vertebrate duplications. *In preparation* (paper 2 of 6 in this series; source in `papers/origin/`).
3. George Dickinson. No vertebrate has lost an IP₃ receptor paralogue. *In preparation* (paper 3 of 6 in this series; source in `papers/retention/`).

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